

AM718 HM Tensile Protocols

Test 1 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 2 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
Specimen had rough surface on gauge region	

Test 3 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN

Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 4 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 5 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
No calibration of extensometer	

Test 6 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 7 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	10 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 8 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm

Additional notes

N/A

AM718 HT1 Tensile Protocols

Test 1 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm

Additional notes

Specimen was not clamped vertically
between grips

Test 2 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm

Additional notes

N/A

Test 3 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN

Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 4 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Croxley ruler with resolution of 0.5 mm
Additional notes	
N/A	

Test 5 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 6 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	10 mm/min
Displacement measurement	Limes video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 7 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limes video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 8 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limes video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm

Additional notes

N/A

AM718 HT2 Protocols

Test 1 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 2 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 3 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN

Strain rate	1 mm/min
Displacement measurement	Crosshead
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 4 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
Specimen fracture outside gauge region	

Test 5 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 6 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 7 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
No calibration of extensometer	

Test 8 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm

Additional notes

N/A

W718 Protocols

Test 1 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	10 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 2 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 3 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN

Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 4 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Crosshead
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 5 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 6 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
N/A	

Test 7 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm
Additional notes	
Specimen fracture outside gauge region	

Test 8 protocol

Universal testing machine	Zwick Roell 1484
Loadcell	100 kN
Strain rate	1 mm/min
Displacement measurement	Limess video extensometer
Measurement of specimen geometry	Vernier callipers with resolution of 0.02 mm

Additional notes

N/A

Hardness Test Protocols

Hardness Test 1 dataset

Test protocol	
Vickers Hardness machine	Zwick Roell ZHV
Load	20 N
Surface preparation	Parallel specimens polished to 3 μm
Number of repeat specimens	30
Additional notes	
Test repeats were performed at least 3 $\times$ diagonal distance of previous indent apart from each other	

Hardness Test 2 dataset

Test protocol	
Vickers Hardness machine	Zwick Roell ZHV
Load	20 N
Surface preparation	Non-parallel specimens polished to 3 μm
Number of repeat specimens	30
Additional notes	
Test repeats were performed at least 3 $\times$ diagonal distance of previous indent apart from each other	

Hardness Test 3 dataset

Test protocol	
Vickers Hardness machine	Zwick Roell ZHV
Load	20 N
Surface preparation	Parallel specimens polished to 3 μm
Number of repeat specimens	30
Additional notes	
Test repeats were performed 1 $\times$ diagonal distance of previous indent apart from each other	

Hardness Test 4 dataset

Test protocol	
Vickers Hardness machine	Zwick Roell ZHV
Load	20 N
Surface preparation	Parallel specimens polished to 3 μm
Number of repeat specimens	15
Additional notes	

Test repeats were performed at least 3× diagonal distance of previous indent apart from each other

KIC Test Protocols

All tests used the following protocol:

Compact tension specimens required for fracture toughness testing was cut out of the rectangular blocks, see Figure below. A specimen width (W) of 32 mm and thickness (B) of 8 mm was used, all other dimensions were chosen according to ASTM specification. K_{IC} measurements were done on specimens pre-cracked to a total crack length (a) of approximately 16 mm corresponding to $a/W \approx 0.5$. The pre-crack extended 4 mm from the tip of the starter notch. Thereafter, it was quasi-statically loaded to failure while measuring load and crack mouth opening displacement. After testing the length of the pre-crack was measured with a traveling microscope so that K_{IC} could be determined.

The samples were tested in the Z-X orientation, meaning the load was applied parallel to the Z-axis and the crack propagated along the X-axis.

